

Enhanced Entity Relationship

ER vs. EER

When we move to an **Enhanced Entity Relationship (EER)**, there are additional concepts:

- **Superclass and subclass**
- **Inheritance**
- **Specialization and generalization**
- **Disjointness - Disjoint vs. Overlapping constraints**
- **Completeness - Total vs. Partial specialization**

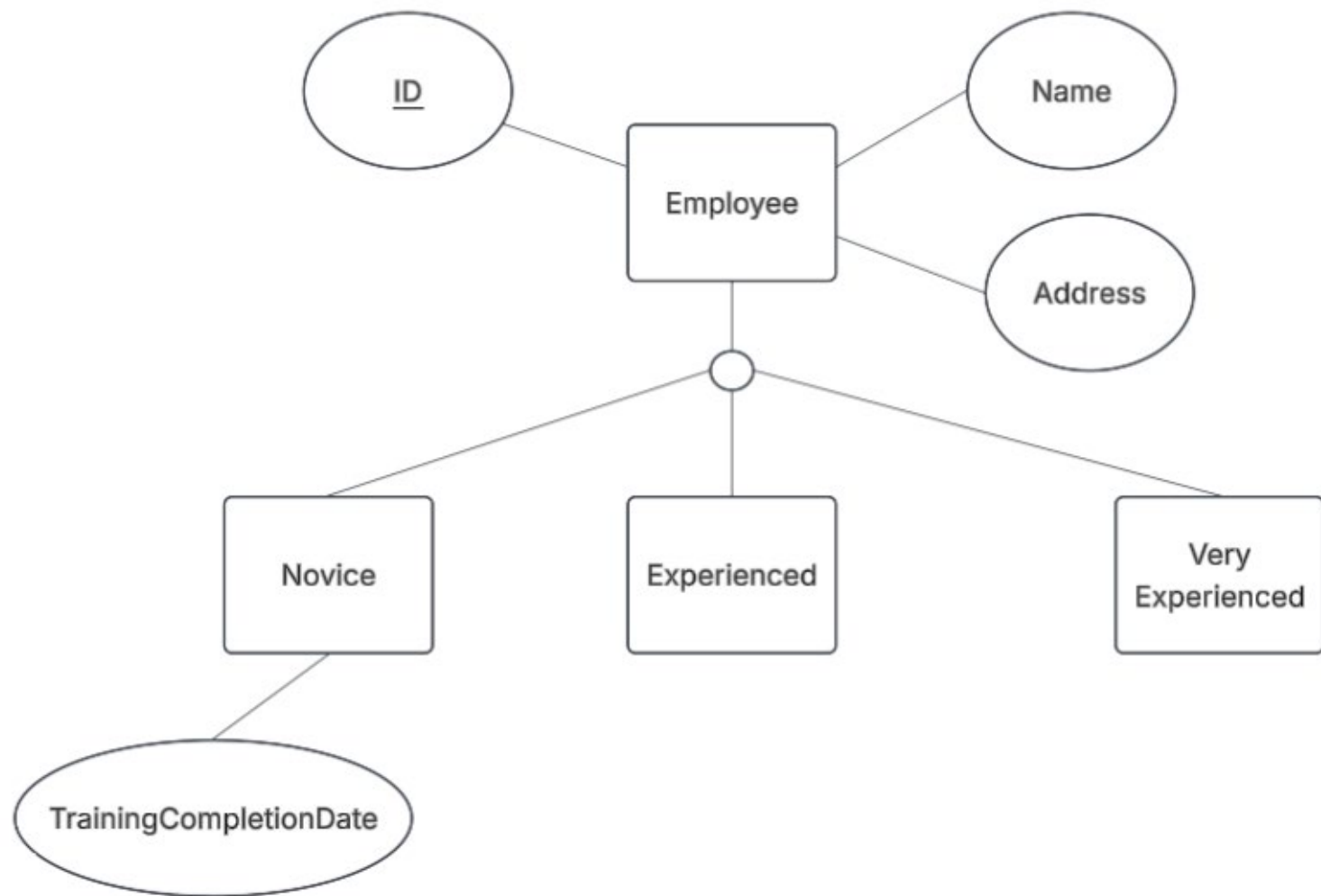
Superclass and Subclass

A **superclass** is a general entity type that contains attributes and relationships common to several more specific entity types. A **subclass** is a specialized entity type that inherits characteristics from the superclass and may have additional attributes or relationships of its own.

Example: Employee -> Novice, Experienced, Very Experienced

A subclass does not need to have attributes of its own. It inherits the attributes and relationships of its superclass and may exist simply to represent different constraints, relationships, or classifications.

A Superclass needs to have at least one attribute.



Inheritance

Inheritance means that a subclass automatically receives the **key attributes, other attributes, and relationships** of its superclass. The subclass can also have its own unique attributes.

Example: Full-Time Employee inherits EmployeeID, Name, and HireDate from Employee.

Specialization and Generalization

- **Specialization** is a **top-down process** that divides a superclass into more specific subclasses based on distinguishing characteristics.
- **Generalization** is a **bottom-up process** that combines similar entity types into a more general superclass.

*Example: Employee → Novice + Experienced + Very Experienced is specialization;
Vehicle → Car + Truck*

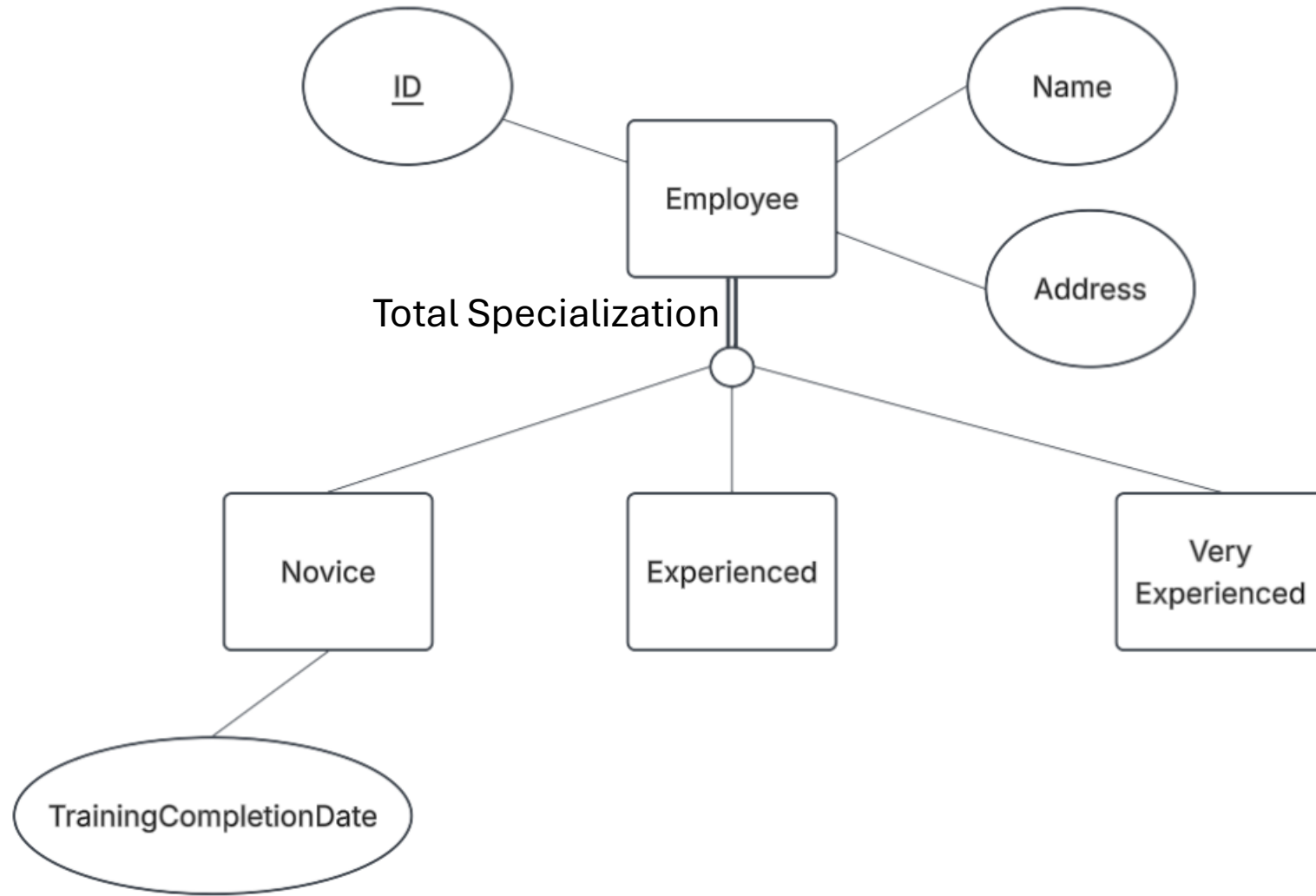
*Novice + Experienced + Very Experienced → Vehicle is a generalization
Car + Truck → Vehicle is generalization.*

Completeness — Total vs. Partial Specialization

The **completeness constraint** determines whether every entity in the superclass must belong to a subclass.

- **Total specialization:** Every superclass entity **must belong to at least one subclass**. Specified by two lines.
- **Partial specialization:** A superclass entity **may exist without belonging to any subclass**. Specified by a single line.

Example: If every Employee must be either Novice, Experienced or Very Experienced, the specialization is Total.



Disjointness — Disjoint vs. Overlapping Constraints

The **disjointness constraint** determines whether a superclass entity may belong to more than one subclass.

- **Disjoint:** An entity can belong to only **one** subclass. Specified by the letter 'D'.

Example: An Employee cannot be both Full-Time and Part-Time = Disjoint.

- **Overlapping:** An entity can belong to **more than one** subclass at the same time. Specified by the letter 'O'.

An University Personnel can be both Faculty and Staff. = Overlapping.

